Glass-Box Green Customer Insights

What we have built since M1

01 Current approach

We help customers and bank staff identify four options: EV finance, green home loans, solar finance and paperless banking. Our approach remains the same as M1. **Models estimate how likely a customer is to choose a product. Rules remove unsuitable options. Calculations estimate environmental benefits, and explanations describe the results.** A high score does not mean a customer qualifies for a loan.

Customer data + grouped transactions → customer profile → product scores
→ product rules → estimated benefits → customer and bank explanations

02 What changed since M1

Since M1, we have added the dataset, trained models, product rules, environmental calculations and Responsible AI checks. We have also brought these components into a FastAPI backend and Streamlit app. Customers can view their own suggestions; bank staff can review customer groups, compare opportunities and try different scenarios.

Component	What is available on main
Data & features	Public data and generated customer records, with one profile per customer. Different customers are used for training, model selection and testing.
Prediction	A model reads transaction descriptions and assigns a category. Four models estimate product uptake and show how confident the system is.
Rules & impact	Rules remove conflicting options. Calculations show the assumptions, conversion factors and units used to estimate benefits. Missing information is checked.
Responsible AI & text	Checks cover data formats, privacy and use of future information. Generated explanations are checked; prepared text is available when generation fails.
Customer & bank app	Separate logins for customers and bank staff. Views show individual suggestions, summaries across customers, model results and what-if scenarios.

03 Dataset and validation results

The main dataset contains **5,000 customers, 276,346 transactions and 20,000 records of product uptake**. We used seed 42 to make generation repeatable. The split is **3,500 training, 750 validation and 750 test customers**. Model inputs use information up to 30 June 2026; records of whether a customer chose a product come from the later outcome period.

The saved validation report shows **all 32 checks passing**. These check record IDs, links between tables, the eight transaction categories, complete labels and separation of customer groups. They also check that future information is excluded from model inputs. The data includes customers with missing information and customers who already own EVs.

Transaction experiment (separate app dataset): TF-IDF turns transaction descriptions into numeric text features; Logistic Regression predicts the category. It scores **0.9996 macro-F1**, versus **0.8358** for keyword rules. Macro-F1 balances incorrect and missed classifications, giving each category equal weight; 1.0 is perfect. Repeated patterns in the generated descriptions make this test easier. Unfamiliar merchants and messier text still need testing.

Current limits: product uptake is generated for the experiment, rules are examples and environmental benefits are estimates. Real bank data and approved lending rules are needed before real-world use.

Sources: data/reports/validation_report.json; product_app/outputs/evaluation_report.json; docs/ENGINE.md; README.md. All results use the main commit shown below.

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Results so far, testing progress and next steps

04 Experiments and results

Product uptake means choosing a product. **LightGBM** combines decision trees; **Logistic Regression (LR)** estimates a probability from weighted customer features. We compared both and their combinations on 750 validation customers per product, choosing the lowest log loss (probability prediction error). We then evaluated 750 test customers per product.
Selected models: EV uses LightGBM only; green home and paperless use 50% LightGBM + 50% LR; solar uses 25% LightGBM + 75% LR.

Product	Val AUROC	Val PR-AUC	Val lift@10%	Test AUROC	LR test AUROC
EV finance	0.574	0.124	1.27	0.578	0.534
Green home	0.586	0.141	1.14	0.638	0.640
Solar finance	0.657	0.172	1.73	0.652	0.647
Paperless	0.655	0.483	1.67	0.658	0.657

How to read the table: Val = validation; LR = Logistic Regression. AUROC measures ranking: 0.5 is random and 1.0 is perfect. PR-AUC balances finding uptake cases against false positives; the code computes it as average precision. Lift@10% compares uptake in the top-scoring 10% with overall uptake: 1.73 means 1.73 times the overall rate.
What the results mean: test AUROC of 0.578–0.658 shows modest ranking ability. The combined models do not always beat LR. Test PR-AUC for EV and home is **0.115 and 0.161**, below LR at **0.137 and 0.176**; LR also gives higher lift for both. The saved report includes uncertainty ranges.

05 Progress on testing

32 / 32 dataset checks PASS in the saved report: IDs, links between tables, categories, labels, separate customer groups and exclusion of future data.

5 model safety tests passed (handoff): existing EV / solar owners and non-homeowners are blocked where required; missing mileage or fuel type triggers review.

6 dataset tests passed (handoff): one profile per customer, four product labels, date limits, separate splits, public input sources and no future labels in model inputs.

105 engine / Responsible AI tests passed (handoff): calculations and units, missing data, response formats, privacy, future-data leakage and unsafe explanation claims.

The reported test suites were not rerun for this report. API (the app's service interface) and login tests also exist, including checks that customers cannot access another customer's records. Full-app verification and live Ollama quality / response-time measurements remain pending.

06 Plan for the remaining time

1. Complete the remaining features. Four of the six planned features are implemented. We plan to add the remaining two before the final submission.

2. Check the integration. Use the same formats and rules across modules. Keep excluded products hidden and emissions, energy, paper and digital measures separate.

3. Test the full app. Rerun the suites and save results for missing data, conflicting products, uncertain scores and access restrictions.

4. Test explanations. Check both audiences for invented claims and numbers. Measure response time and confirm prepared text appears when generation fails.

5. Finish the submission. Test harder transaction examples, label each result's dataset and prepare the final report, setup instructions, video and code archive.

Sources: ml/outputs/project_evaluation_report.json; product_app/SUBMISSION.md; ml/tests/test_safety.py; tests/test_dataset.py; tests/engine/; tests/responsible_ai/; product_app/tests/.